EINDHOVEN UNIVERSITY OF TECHNOLOGY

(2MMN40) Introduction to Molecular Modeling and Simulation

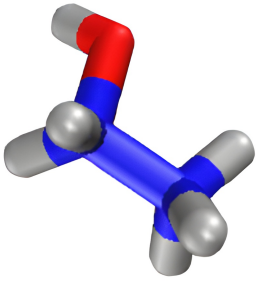
Final Project

Simulating Water and Ethanol



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1 Abstract

This report presents a molecular dynamics simulation study of water, ethanol, and their mixtures, with a focus on both inter- and intramolecular interactions. The simulation models three systems: pure water, pure ethanol, and a 14.3% ethanol-in-water mixture, using various force fields and numerical integration methods. Key intermolecular forces, including bond, angular, dihedral, and Lennard-Jones potentials, are implemented to capture molecular interactions. The simulations analyze structural properties through radial distribution functions and energy conservation, with results compared to literature data.

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2 Introduction

The goal of this paper is to simulate the following systems:

- A system of pure water;
- A system of pure ethanol;
- A system of 14.3% ethanol in water (percents are in mole).

An overview of all parameters and symbols used in this report can be found in the appendix in A.1.

2.1 Molecules with parameters

This paper delves into the interactions between water- and ethanol molecules. This paragraph covers their structure and properties. The atoms found in these molecules are:

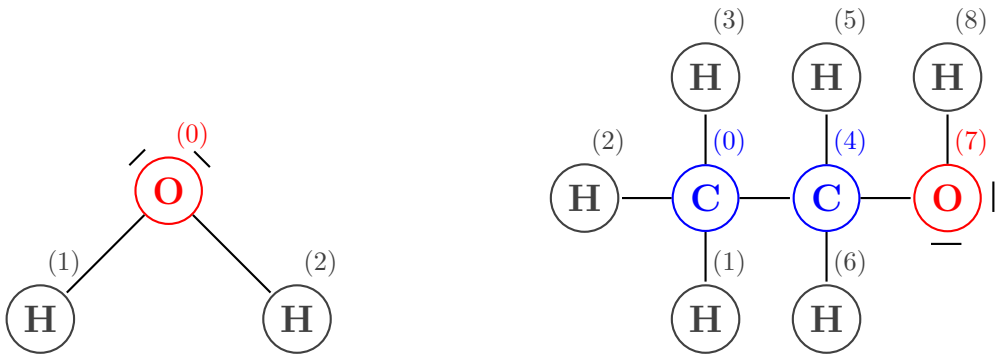
- Hydrogen (**H**);
- Oxygen (**O**);
- Carbon (**C**).

2.1.1 Water

A water molecule consists of one oxygen (**O**) and two hydrogen (**H**) atoms. The outer shell of oxygen has six electrons and it needs eight electrons. Therefore there are two electron pairs around oxygen as well as the two bonds with both hydrogen atoms, creating an angular shape in the molecule's structure (more on this in subsection 2.2.2). Hydrogen has one electron in its outer shell and needs two. This leads to the structure in Figure 2.1a.
The parameters used to describe water are listed in section B.1.

2.1.2 Ethanol

An ethanol molecule consists of a core structure of two carbon (**C**) atoms and one oxygen (**O**) atom with all shells filled using hydrogen (**H**) bonds. The **O** atom also has two electron pairs like water. Carbon needs four bonds (so eight electrons) in total. This leads to the structure in Figure 2.1b.
The parameters used to describe ethanol are listed in section B.2.



(a) Water (b) Ethanol

Figure 2.1: Structures of the molecules

2.2 Intramolecular forces

In this section, we look into the different forces that are affecting the acceleration of each atom within a molecule. There are three forces taken into account in this report:

- the bond force;
- the angular force;
- the dihedral force.

We look at the force between different molecules in the next section.

2.2.1 Bond force

Some electrons are shared between two atoms to achieve a stable electron configuration. An example of this would be the electron pair between **H** and **O** in water. This gives rise to a force that keeps these nuclei together. The attractive force comes from the magnetic force between the negative electron pair and both positive nuclei. The repulsive force arises when the positive nuclei are coming too close to each other. We can approximate the potential V_{bond} of this behavior using a function that is quadratic in r, the distance between the atoms:

$$V_{\text{bond}}(r) = \frac{1}{2} k_b (r - r_0)^2 \tag{2.2.1}$$

where

- r_0 is the most energetically favorite distance between the atoms;
- k_b is the strength of the bond.

Both parameters depend on which two atoms are bonded. The values that we use, are given in Table B.2 for water and in Table B.5 for ethanol.

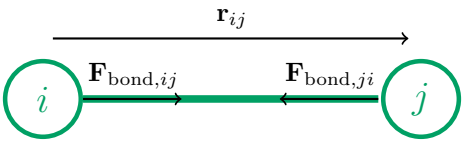


Figure 2.2: Bond force illustration

We find that the bond force of one pair can be determined as follows:

$$\mathbf{F}_{\text{bond},ij} = k_b(r_{ij} - r_0) \cdot \frac{\mathbf{r}_{ij}}{|\mathbf{r}_{ij}|} \qquad \mathbf{F}_{\text{bond},ji} = -\mathbf{F}_{\text{bond},ij}$$

For a full calculation of this force, see appendix A.2.1.

2.2.2 Angular force

Most atoms need to have eight electrons, so four pairs of electrons, in the outer shell. This means that there are four negatively charged groups that want to be close to the positive nucleus, yet want to be as far away as possible from each other. This means that there will be a tetrahedron structure in water. A visualization of this structure is given in Figure 2.3. The same will happen in ethanol around the **C** and **O** atoms. We can see that there will be a force for each pair away from each other pair. The magnitude of this force depends on the distance between the pairs, as well as the angle. The sum of these forces for one pair is what we define here as the angle force for that pair. Each triple in both water and ethanol

has its own equilibrium angle. We can then approximate the potential $V_{\text{angle}}$of this behavior by using a function quadratic in the angle θ:

$$V_{\text{angle}}(\theta) = \frac{1}{2}k_\theta(\theta - \theta_0)^2 \tag{2.2.2}$$

where

- θ_0 is the most energetically favorite angle between the atoms;
- k_θ is the strength of the angle.

Both parameters depend on which three atoms are making the angle. The values that we use are given in Table B.3 for water, and in Table B.6 for ethanol. For further definition of i, j and k, we use the triple visualized in Figure 2.4. The angular force on i in angle $\angle ijk$ is defined as $\mathbf{F}_{\text{angle},i,ijk}$. We also define this force on j as $\mathbf{F}_{\text{angle},j,ijk}$ and on k as $\mathbf{F}_{\text{angle},k,ijk}$.

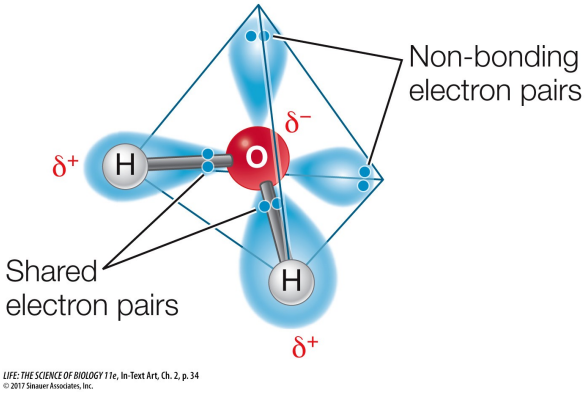


Figure 2.3: Tetrahedron structure of water. [1]

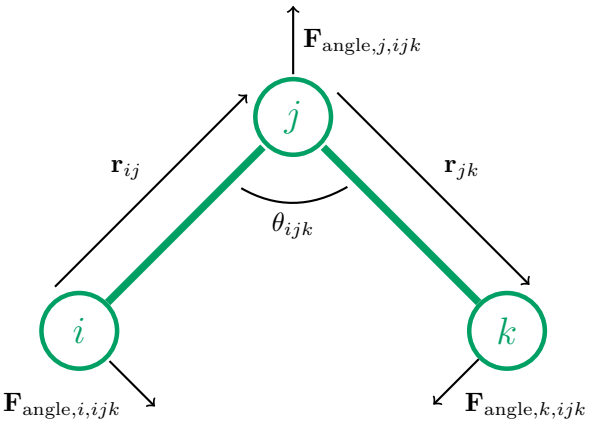


Figure 2.4: Angle force illustration.

From this, the angle force of one triple is found to be:

$$\begin{aligned}
\mathbf{F}_{\text{angle},i,ijk} &= \frac{k_b}{|\mathbf{r}_{ij}|}(\theta_0 - \theta_{ijk})\frac{\mathbf{r}_{ij} \times \hat{\mathbf{n}}_{ijk}}{|\mathbf{r}_{ij} \times \hat{\mathbf{n}}_{ijk}|} \\
\mathbf{F}_{\text{angle},k,kji} &= \frac{k_b}{|\mathbf{r}_{kj}|}(\theta_0 - \theta_{kji})\frac{\mathbf{r}_{kj} \times \hat{\mathbf{n}}_{kji}}{|\mathbf{r}_{kj} \times \hat{\mathbf{n}}_{kji}|} \\
\mathbf{F}_{\text{angle},j,ijk} &= -\left(\mathbf{F}_{\text{angle},i,ijk} + \mathbf{F}_{\text{angle},k,ijk}\right)
\end{aligned}$$

For a full calculation of this force, see appendix A.2.2.

2.2.3 Dihedral force

For any two bonded atoms, all surrounding bonded atoms have a prefered rotation relative to each other. This is a consequence of magnetic forces caused by potential electron pairs and the cores. The preference can either be close to each other (cis) or as far away as possible to each other (trans). We can encapsulate this behavior in the Ryckaert-Bellman potential:

$$V_{\text{dihedral}}(\psi) = \frac{1}{2}\Big[C_1(1 + \cos\psi) + C_2(1 - \cos 2\psi) + C_3(1 + \cos 3\psi) + C_4(1 - \cos 4\psi)\Big] \tag{2.2.3}$$

where

- $\psi = \psi_{ijkl} - 180°$ is the angle between the ijk-plane and the jkl-plane;

- $C_1, \ldots, C_4$ are given constants. All parameters depend on which four atoms are interacting. The values that we use, are given in Table B.7 for ethanol.

For further definition of i, j, k and l, we use the quadruple visualized in Figure 2.5. We define the dihedral force on i in plane angle ψ_{ijkl} as $\mathbf{F}_{\text{dihedral},i,ijkl}$, and similarly for j, k and l.

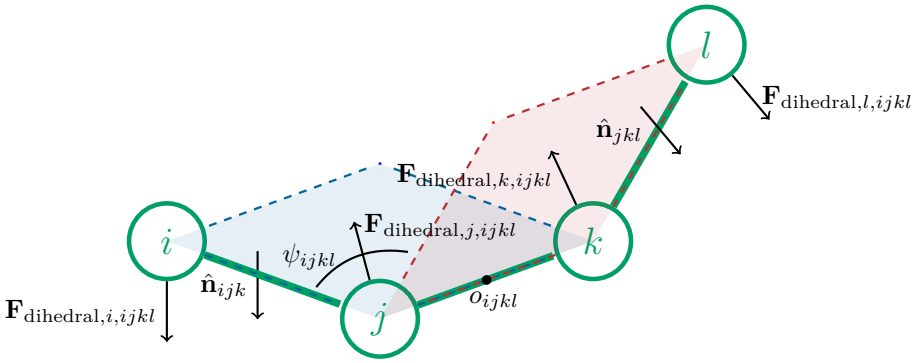


Figure 2.5: Dihedral force illustration.

We find that the dihedral force of one quadruple can be determined as follows:

$$\mathbf{F}_{\text{dihedral},i,ijkl} = \frac{1}{2|\mathbf{r}_{ij}|\sin\theta_{ijk}}\Big[C_1\sin\psi - 2C_2\sin 2\psi + 3C_3\sin 3\psi - 4C_4\sin 4\psi\Big]\hat{\mathbf{n}}_{ijk}$$

$$\mathbf{F}_{\text{dihedral},l,ijkl} = \frac{1}{2|\mathbf{r}_{lk}|\sin\theta_{lkj}}\Big[C_1\sin\psi - 2C_2\sin 2\psi + 3C_3\sin 3\psi - 4C_4\sin 4\psi\Big]\hat{\mathbf{n}}_{lkj}$$

$$\mathbf{F}_{\text{dihedral},k,ijkl} = \left(\frac{1}{|\mathbf{r}_{ol}|^2}\right)\mathbf{t}_k \times \mathbf{r}_{ok}$$

$$\mathbf{F}_{\text{dihedral},j,ijkl} = -\mathbf{F}_{\text{dihedral},i,ijkl} - \mathbf{F}_{\text{dihedral},k,ijkl} - \mathbf{F}_{\text{dihedral},l,ijkl}$$

where point o was introduced to be the center of point j and k and

$$\mathbf{t}_k := -\left(\mathbf{r}_{ok} \times \mathbf{F}_{\text{dihedral},l,ijkl} + \frac{1}{2}\mathbf{r}_{kl} \times \mathbf{F}_{\text{dihedral},l,ijkl} + \frac{1}{2}\mathbf{r}_{ji} \times \mathbf{F}_{\text{dihedral},i,ijkl}\right)$$

For a full calculation of this force, see appendix A.2.3.

2.3 Intermolecular forces

For this project, we only look at the Lennard Jones potential as an intermolecular force. This potential models the interaction between two atoms (labeled i and j) that are in different molecules. The potential is defined as:

$$U_{ij}^{LJ}(r) = 4\epsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r}\right)^{12} - \left(\frac{\sigma_{ij}}{r}\right)^{6}\right] \tag{2.3.1}$$

where

- σ_{ij} is the distance at which the potential is zero (effective particle diameter);
- ϵ_{ij} is the depth of the potential well (energy scale).

Both constants depend on which two atoms are interacting. We determine the value for the pair ij by looking at the individual σ and ϵ values of i and j and using the following 'mixing rules':

$$\sigma_{ij} = \frac{1}{2}(\sigma_i + \sigma_j) \qquad \epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j} \tag{2.3.2}$$

The values that we use for $\sigma_i, \sigma_j, \epsilon_i$ and ϵ_j, are given in Table B.1 for water and in Table B.4 for ethanol. The mixing rules above are based on the Lorentz-Berthelot rules. Lorentz found that an arithmetic mean is fitting for σ_{ij} [2] and Berthelot found a geometric mean [3] for ϵ_{ij}. This implies that the potential is zero if at least one of the ϵ_i is zero, meaning that we do not look at forces between such atoms (which saves time spent on on these computation). Many other mixing rules exist, like those of Waldman-Hagler, Fender-Halsey or Kong.

The potential consists of a repulsive and attractive component:

- The *repulsive force* can be explained with the Pauli exclusion principle. When two atoms come very close, their electron clouds overlap. No two electrons with the same quantum state can occupy the same space, therefore leading to strong repulsion at short distances. This behavior can be estimated with a term that is $\mathcal{O}(r^{-12})$;
- The *attractive force* is caused by van der Waals dispersion forces. This force arises due to dipole interactions. An example would be that the **H** side of water is a little bit positively charged and the **O** side is a little bit negatively charged. This behavior can be estimated with a term that is $\mathcal{O}(r^{-6})$.

Now combining these two factors (and scaling with the mixed parameters), we find the potential in Equation 2.3.1. We define the Lennard Jones force on particle i induced by particle j as $\mathbf{F}_{\mathrm{LJ},ij}$.



Figure 2.6: Lennard Jones force illustration on two water molecules.

We find that the Lennard-Jones force of one pair can be determined as follows:

$$\mathbf{F}_{\mathrm{LJ},ij} = \frac{24\epsilon_{ij}}{r_{ij}} \left[2\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6} \right] \cdot \frac{\mathbf{r}_{ij}}{|\mathbf{r}_{ij}|}$$

$$\mathbf{F}_{\mathrm{LJ},ji} = -\mathbf{F}_{\mathrm{LJ},ij}$$

For a full calculation of this force, see appendix A.3.

2.4 Previous studies

Ethanol-water mixtures have been widely studied using molecular dynamics simulations (MDS). Some interesting phenomena to look at are:

- Hydrogen bonding and structural organization: Water forms a hydrogen bond network which is disrupted by ethanol. MDS have shown clustering of ethanol at lower concentrations. These water-rich and ethanol-rich regions can be observed using spectroscopy or X-ray scattering techniques. [4] [5]
- Thermodynamic and dynamic properties: MDS reveal that ethanol slows down water diffusion and induces deviations in entropy and enthalpy. MDS provide insights into this macroscopic phenomenon. [6]
- Influence of different mol percentages: O. Gereben and L. Pusztai looked at varying mol percentages using the OPLS force field and three different water models to analyse structural properties [7]. D.

Benavides Bautista, M. Aguilar and O. Pizio did research on various properties of water-ethanol mixtures. The study focused on composition trends in thermodynamic properties at constant pressure and temperature [8].

To approximate the molecular behavior of the mixture, various force fields have been determined, like the TIP4P model for water [9] and the OPLS-AA for ethanol [10]. The code developed in this project should be able to show these phenomena, provided the parameters (like the number of time steps and -molecules) are set appropriately.

3 Method

Figure 3.1 shows a flowchart that gives a clear view of the code and it's main components. More details are given in the sections below.

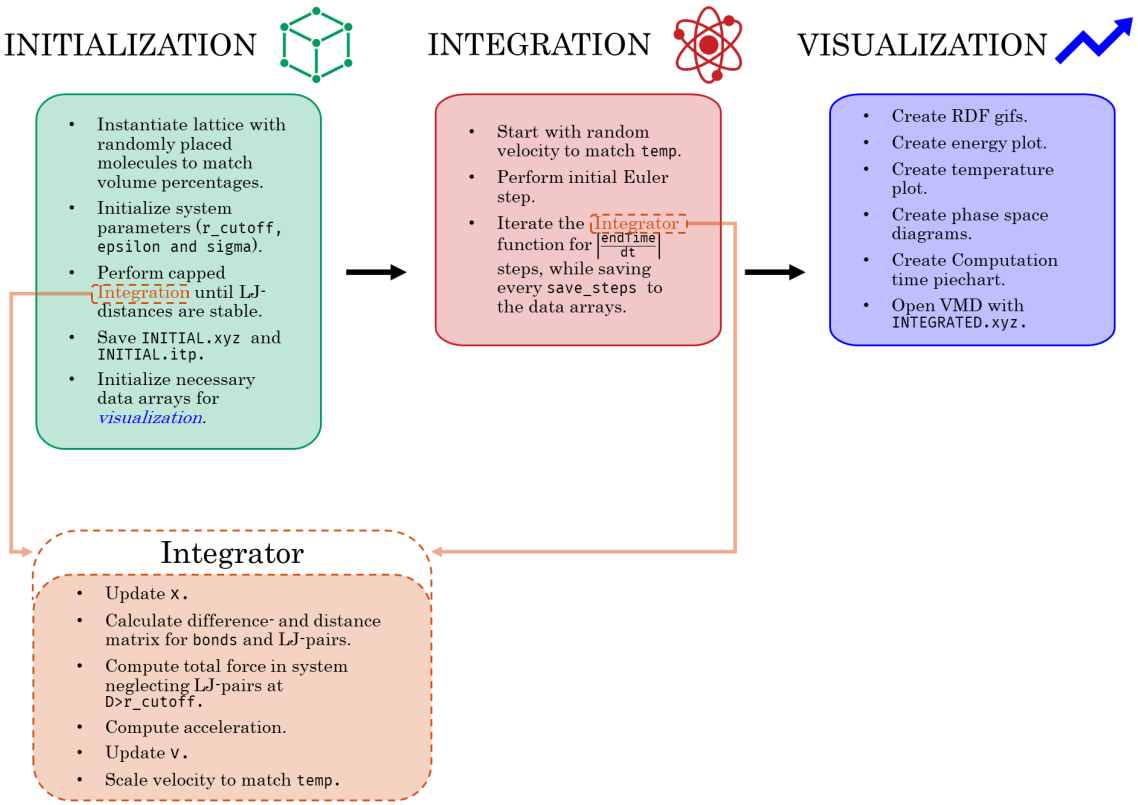


Figure 3.1: Flowchart model of the code

3.1 Initialization

To initialize the system, we first compute the number of molecules of each type that is needed to achieve the molecular volume in the bounding box. We are given percentages per molecule in the vector w, which will typically look something like $(0.857, 0.143, 0, 0)$, where each entry corresponds to the percentages of water, ethanol, oxygen and hydrogen in the system. Note that this model allows for simulations of more than just water and ethanol, although it will not be covered in this report.

From w, we determine the the number of molecules of each type by computing

$$B^3 \cdot N_A \cdot \sum_{i=1}^{4} \frac{w_i}{\tilde{V}_i}$$

In section 3.1, B denotes the boxsize in which the simulation takes place (in this report 3.5nm), N_A represents Avogadro's number $\approx 6.02 \cdot 10^{23}$ and $\tilde{V}_i$ stands for the molar volume of each molecule.

Once the number of molecules per type is determined, their center points are placed on randomly chosen positions in the smallest cubic three-dimensional lattice that contains them. The molecules are rotated to fit in a cube-like bounding box, as to minimize the distance between two atoms in neighbouring molecules.

Note that some positions on this lattice may be left open, since the total number of molecules may not necessarily be a cubed integer.

After the molecules are placed, the model parameters are created by repeating the data in the ITP files of the original molecules. After this, some simulation parameters are computed, including the identification of LJ-pairs and their corresponding parameters in `epsilon`, `sigma` and `r_cutoff`. This is done by determining which pairs of atoms (i, j) do not interact through dihedral-, angular- or bond forces, and identifying pairs that have a shared product $\sqrt{\epsilon_i \epsilon_j} > 0$ and $\frac{1}{2}(\sigma_i + \sigma_j)$. In the systems we are simulating, this boils down to not simulating LJ-forces within a single molecule. We chose to not use a constant value for `r_cutoff`, but one that linearly scales with the equalibrium distance (it is written as $c \cdot \sigma$ for a scalar c). In literature, c is often chosen as 2 or 2.5, the former of which we use too.

After this, the simulation is almost ready to run; all that remains is to integrate the system for a few steps with bounded LJ-forces, until the LJ-pairs have reached a threshold distance value. This step is necessary to prevent any extreme forces caused by low distances in the initial configuration.

After the intial configuration has settled, it's position data is saved to a file, and the integration process begins.

3.2 Integration

The integration process allows for a choice between four integrator methods, each updating the position and velocity of the system:

- Euler's Algorithm;
- Verlet's Algorithm;
- Velocity Verlet's Algorithm;
- Beeman's Algorithm.

The integration process is quite straightforward; in each iteration, the following steps are executed:

- First, the positions are updated according to the chosen integrators position update component.
- Then, the differences and distances between bonds and LJ-pairs are computed using the function called `diff_and_dist_arr`. These distances are used for intra- and intermolecular compuatations respectively. The `diff_and_dist_arr` function is written to respect Periodic Boundary Conditions (PBC), meaning that the system is seen as an infinite three dimensional tiling of the particles found in the box. To achieve this, any difference between a pair (i, j) is modulated by the boxsize B in each axis, mapping the resulting vector to be in $[-\frac{B}{2}, \frac{B}{2})$. This modulation is chosen such that each particle interacts only with its closest counterparts, since no difference can exceed $\frac{B}{2}$ in absolute value for any of its components.
- After this, the forces are computed using the `net_force` function (which sums the results of four other functions). These functions can be found in section 2.2 and section 2.3.

- The acceleration is computed from the resulting force, using the masses of each particle.
- Finally, the velocities are updated again using the quantities above. After the velocity update has been computed, the velocity is scaled to match the provided temperature `temp` by scaling it with a factor $\sqrt{\frac{T \cdot N_f \cdot k_b}{2E_{\text{kin}}}}$ This scaling ensures that the temperature remains constant over time, which is an accurate assumption for the timescale we are simulating. In the above, E_{kin} denotes the kinetic energy (which is determined by the velocities), N_f is the number of degrees of freedom in the system, k_b is Boltzmann constant and T denotes temperature. This scaling of velocities is common practice in molecular simulations, and is often referred to as a *thermostat*.

In addition, the program saves data to a file every `save_steps` steps, including the positions, computation times, energy and distances. More on this in section 3.3. Note that at no point do we ensure the particles remain in the box; they are in fact allowed to move outside, but will interact with other particles as if they were in the box together. It is only during the data-saving process that we project the particles back into the box, for easier viewing.

The computational complexity of this process can be estimated as follows: doubling the box size results in approximately 2^3 copies of the same system. Since the distance between each pair of particles must be analyzed, which requires quadratic time, the complexity for a single timestep is $\mathcal{O}(B^6)$. It is evident that the total complexity scales linearly with the number of timesteps, which is given by $\frac{\texttt{endTime}}{\texttt{dt}}$. Therefore, neglecting the small contribution from saving data, the total time complexity of the full simulation is

$$\mathcal{O}\left(\frac{\texttt{endTime}}{\texttt{dt}} \cdot B^6\right). \tag{3.2.1}$$

3.2.1 Integrators

The integration methods use the following Taylor expansions:

$$\mathbf{q}_i(t+\Delta t) = \mathbf{q}_i(t) + \Delta t \dot{\mathbf{q}}_i(t) + \frac{\Delta t^2}{2}\ddot{\mathbf{q}}_i(t) + \frac{\Delta t^3}{3!}\dddot{\mathbf{q}}_i(t) + \mathcal{O}(\Delta t^4) \tag{3.2.2}$$

$$\mathbf{p}_i(t+\Delta t) = \mathbf{p}_i(t) + \Delta t \dot{\mathbf{p}}_i(t) + \frac{\Delta t^2}{2}\ddot{\mathbf{p}}_i(t) + \frac{\Delta t^3}{3!}\dddot{\mathbf{p}}_i(t) + \mathcal{O}(\Delta t^4) \tag{3.2.3}$$

$$\mathbf{v}_i(t+\Delta t) = \mathbf{v}_i(t) + \Delta t \dot{\mathbf{v}}_i(t) + \frac{\Delta t^2}{2}\ddot{\mathbf{v}}_i(t) + \frac{\Delta t^3}{3!}\dddot{\mathbf{v}}_i(t) + \mathcal{O}(\Delta t^4) \tag{3.2.4}$$

with

$$\mathbf{a}_i(t) = \dot{\mathbf{v}}_i(t) = \ddot{\mathbf{q}}_i(t) = \frac{\mathbf{F}_i(t)}{m_i} \qquad \mathbf{p}_i(t) = m_i \mathbf{v}_i(t) \tag{3.2.5}$$

where

- $\mathbf{q}_i$ is the position of atom i;
- $\mathbf{p}_i$ is the momentum of atom i;
- $\mathbf{v}_i$ is the velocity of atom i;
- t is the current time;
- Δt is the time step;
- $\mathbf{F}_i$ is a force on atom i;
- m_i is the mass of atom i;
- $\mathbf{a}_i$ is the acceleration of atom i.

We take a closer look on these algorithms in this section.

3.2.1.1 Euler's Algorithm

This algorithm uses the forward difference scheme in $\mathbf{v}$ and is second order in $\mathbf{q}$ and first order in $\mathbf{v}$:

$$\mathbf{q}_i(t+\Delta t) = \mathbf{q}_i(t) + \Delta t \mathbf{v}_i(t) + \frac{\Delta t^2}{2}\mathbf{a}_i(t) \tag{3.2.6}$$

$$\mathbf{v}_i(t+\Delta t) = \mathbf{v}_i(t) + \Delta t \mathbf{a}_i(t) \tag{3.2.7}$$

This algorithm does not work perfectly as it does not have time-reversibility and no phase space conservation.

The python implementation of this integrator can be found in C.1.1.

3.2.1.2 Verlet's Algorithm

Verlet's Algorithm takes Equation 3.2.2 to the second order and we also consider the version where we change the sign of Δt:

$$\mathbf{q}_i(t+\Delta t) = \mathbf{q}_i(t) + \Delta t \mathbf{v}_i(t) + \frac{\Delta t^2}{2}\mathbf{a}_i(t) \tag{3.2.8}$$

$$\mathbf{q}_i(t-\Delta t) = \mathbf{q}_i(t) - \Delta t \mathbf{v}_i(t) + \frac{\Delta t^2}{2}\mathbf{a}_i(t) \tag{3.2.9}$$

Now we both add and substract the two:

$$\mathbf{q}_i(t+\Delta t) + \mathbf{q}_i(t-\Delta t) = 2\mathbf{q}_i(t) + \Delta t^2 \mathbf{a}_i(t) \tag{3.2.10}$$

$$\mathbf{q}_i(t+\Delta t) - \mathbf{q}_i(t-\Delta t) = 2\Delta t \mathbf{v}_i(t) \tag{3.2.11}$$

We now find the position update using Equation 3.2.10:

$$\mathbf{q}_i(t+\Delta t) = 2\mathbf{q}_i(t) - \mathbf{q}_i(t-\Delta t) + \Delta t^2 \mathbf{a}_i(t) \tag{3.2.12}$$

and the velocity update using Equation 3.2.11:

$$\mathbf{v}_i(t+\Delta t) = \frac{\mathbf{q}_i(t+\Delta t) - \mathbf{q}_i(t-\Delta t)}{2\Delta t} \tag{3.2.13}$$

The python implementation of this integrator can be found in C.1.2.

3.2.1.3 Velocity Verlet's Algorithm

For this integrator, we take Equation 3.2.2 and Equation 3.2.4 to the second order and we then rewrite the first derivative of $\mathbf{v}_i(t)$ to first order Taylor expansion:

$$\dot{\mathbf{v}}_i(t+\Delta t) = \dot{\mathbf{v}}_i(t) + \Delta t \ddot{\mathbf{v}}_i(t) \tag{3.2.14}$$

$$\Delta t \ddot{\mathbf{v}}_i(t) = \dot{\mathbf{v}}_i(t+\Delta t) - \dot{\mathbf{v}}_i(t) \tag{3.2.15}$$

$$\frac{(\Delta t)^2}{2}\ddot{\mathbf{v}}_i(t) = \frac{\Delta t}{2}\dot{\mathbf{v}}_i(t+\Delta t) - \frac{\Delta t}{2}\dot{\mathbf{v}}_i(t) \tag{3.2.16}$$

$$= \frac{\Delta t}{2}\left(\mathbf{a}_i(t+\Delta t) - \mathbf{a}_i(t)\right) \tag{3.2.17}$$

We now substitude this in the second order version of Equation 3.2.4 to find the final form:

$$\mathbf{q}_i(t+\Delta t) = \mathbf{q}_i(t) + \Delta t \mathbf{v}_i(t) + \frac{\Delta t^2}{2}\mathbf{a}_i(t) \tag{3.2.18}$$

$$\mathbf{v}_i(t+\Delta t) = \mathbf{v}_i(t) + \frac{\Delta t}{2}\left(\mathbf{a}_i(t) + \mathbf{a}_i(t+\Delta t)\right) \tag{3.2.19}$$

This integrator is considered really good to use for the simulation. It is accurate enough yet relatively easy to compute as it is the second order Taylor approximation. It also has time reversibility. *The python implementation of this integrator can be found in C.1.3.*

3.2.1.4 Beeman's Algorithm

The last integrator is based on Beeman's algorithm. We take Equation 3.2.2 to the third order and Equation 3.2.4 to the second order. Then we approximate $\dot{\mathbf{a}}_i(t)$ using the backward difference method:

$$\dot{\mathbf{a}}_i(t+\Delta t) = \frac{\mathbf{a}_i(t) - \mathbf{a}_i(t-\Delta t)}{\Delta t} \tag{3.2.20}$$

$$\mathbf{q}_i(t+\Delta t) = \mathbf{q}_i(t) + \Delta t \mathbf{v}_i(t) + \frac{\Delta t^2}{2}\mathbf{a}_i(t) + \frac{\Delta t^3}{6}\dot{\mathbf{a}}_i(t) \tag{3.2.21}$$

$$= \mathbf{q}_i(t) + \Delta t \mathbf{v}_i(t) + \frac{\Delta t^2}{2}\mathbf{a}_i(t) + \frac{\Delta t^3}{6}\left(\frac{\mathbf{a}_i(t) - \mathbf{a}_i(t-\Delta t)}{\Delta t}\right) \tag{3.2.22}$$

To find the position update and said velocity update:

$$\mathbf{q}_i(t+\Delta t) = \mathbf{q}_i(t) + \Delta t \mathbf{v}_i(t) + \frac{\Delta t^2}{6}\left(4\mathbf{a}_i(t) - \mathbf{a}_i(t-\Delta t)\right) \tag{3.2.23}$$

$$\mathbf{v}_i(t+\Delta t) = \mathbf{v}_i(t) + \frac{\Delta t}{6}\left(2\mathbf{a}_i(t+\Delta t)\mathbf{a}_i(t) - \mathbf{a}_i(t-\Delta t)\right) \tag{3.2.24}$$

This integrator is a bit better than Velocity Verlet's Algorithm, but it also requires a bit more computation time as it is a third order Taylor approximation. *The python implementation of this integrator can be found in C.1.4.*

3.3 Visualization

After the integration process is complete, the program creates some plots and graphs of interesting data to analyze the simulation:

3.3.1 Radial Distribution Function

The radial distribution function g_{AB} is a density function that shows the probability of finding an atom of type B at a certain distance from A. This function can be estimated by measuring these distances over time, and plotting histogram(s).

In literature, the radial distribution is often presented as a single figure, containing time averaged data of the distances. In this paper, we aim to show the evolution of this radial distribution function over time.

3.3.2 Energy

In any isolated physical model, total energy is preserved. Thus, we hope to see that the total energy in the system (which consists of bond potential, angular potential, kinetic energy, dihedral potential and LJ-potential) remains constant over time. However, since this simulation is *temperature-controlled*, we will not be observing a total-energy in the system; the kinetic energy will be scaled to a constant value. Thus, this figure will only provide useful insights in the dynamics of the system, and should not be expected to have constant total energy. The functions used to determine the systems energy can be found in section 2.2 and section 2.3.

3.3.3 Phase space diagram

For each bond present in the simulation, we can track its distance over time and show how this evolves in a *phase space diagram*; this diagram plots the distance of all bonds of a given type against their velocities,

creating a trajectory. These figures often give rise to elliptical shapes, which corresponds to the known solutions to spring-dynamic systems that take on a sinusoidal form.

4 Results

Due to space constraints, this section will only show the highlights of the results found in this project. To see all of the results (including GIFS and trajectory files), one may visit the repository [11]. The highlights of the results of a system of pure water and pure ethanol can be found in the appendices D.1 and D.2 respectively.
All simulations were run with the following parameters:

```
# Variables related to weights and box size
boxsize = 3.5
random_grid_offset = 0
r_threshold_scalar = 0.65

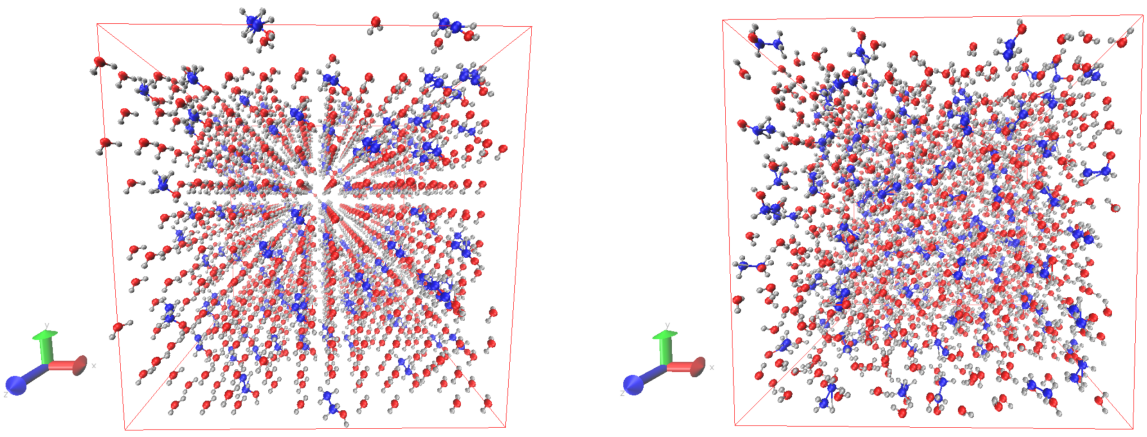
# Integration parameters
dt = 0.001
endTime = 0.5
save_steps = 100
method = velo_verlet
initial_temp = 298.15
temp_control = True

# Force thresholds and cutoffs
f_threshold = None
r_cutoff_scalar = 2

# Visualization parameters
RDF_thin = 0.1
RDF_dr = 0.05
```

4.1 Mixed system

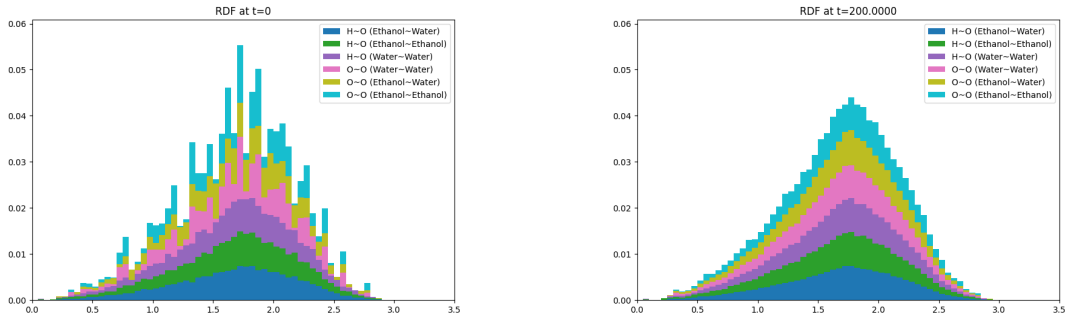
For this system, we let `weights=[0.857,0.143,0,0]`. The XYZ file for this simulation was found on [12]. Running the simulation provides the following figures:



(a) Initial configuration

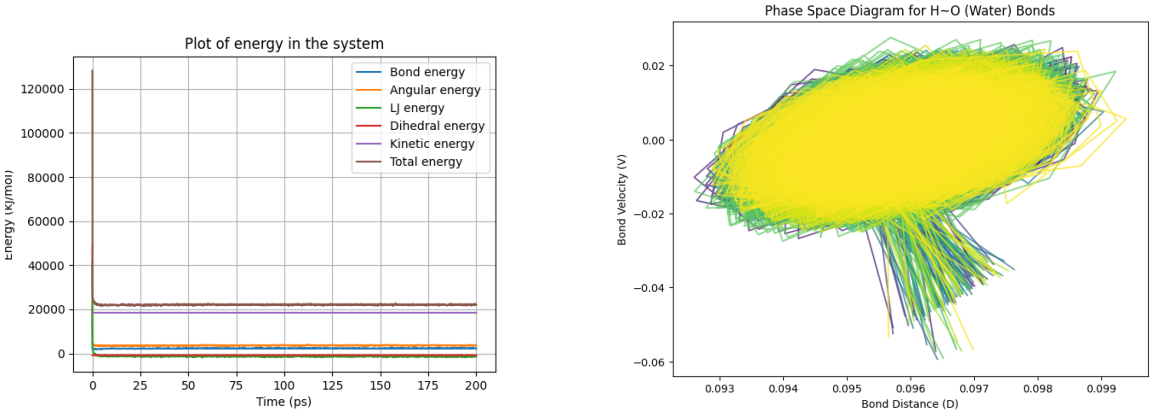
(b) Later configuration

Figure 4.1: Two frames of the simulation, visualized with VMD



(a) at $t = 0$ ps (b) at $t = 200$ ps

Figure 4.2: Radial distribution of the **H**-**O** and **O**-**O** bonds. x-axis is in nm.



(a) Energy of the system over time (b) Phase space diagram for **H**-**O** bonds

Figure 4.3: Two diagrams of interest.

5 Discussion & Conclusion

5.1 Comparison to literature

We compare our radial distribution function (RDF) with the graph presented by C. Zhang and X. Yang in 2005 [13]. As noted in subsection 3.3.1, their approach to visualization differs from ours. Specifically, their plot is scaled by distance to enhance the identification of peaks. While these peaks are also observable in our results (e.g., around $r = 0.035$), they are less prominent due to the curvature inherent in our histogram. A scaled version of our plot would similarly highlight these peaks, improving their visibility.

The presence of these peaks suggests that specific atom pairs in the system exhibit a preference for smaller distances, indicative of clustering behavior. Notably, the simulations demonstrate that both water and ethanol molecules favor configurations in which they cluster together.

In addition, the phase space diagrams serve as an effective sanity check, confirming that bond oscillations occur in a natural and expected manner, with all observed shapes appearing predominantly elliptical. The jagged edges and trajectories through the center point are consistent with artifacts arising from data being saved at discrete intervals. The small "stem" visible at the bottom of the ellipse in Figure 4.3b can

be attributed to the initial step of the integration process, which may introduce velocities in non-physical directions for the bonds. This minor anomaly is quickly corrected by the integrator in the following steps.

5.2 Reflection

This project provided us with valuable insights into molecular modeling, encompassing both theoretical principles and their efficient implementation. We were very glad to also be taking the course scientific programming in parallel to this, which enabled us to apply techniques learned there to optimize our simulation.

In general, we are quite content with the flexibility of our code. It works not only with ethanol and water, but any molecule (given its xyz and itp data). The initialization process works quite well, and makes it easy to input volume percentages for each molecule type. The feature to save data only every few steps saved a lot of computation time, and the option to manually toggle what bonds/pairs to visualize in the phase space diagrams and RDF's are very user-friendly.

We are also quite content with our implementation of LJ-interactions, even allowing for different values of `r_cutoff` depending on the pair. This implemntation allowed us to simulate the systems in this report in reasonable time (the ethanol system took a little more than two days). However, the code falls short when applied to larger systems or longer simulation times, making it difficult to observe phenomena on larger scales. Unfortunately, we were unable to simulate the suggested systems within the timeframe specified in the assignment. In our assessment, we believe we have optimized the code as much as possible within the constraints of the current implementation. Further performance improvements would likely require transitioning to a more efficient programming language, as we appear to have reached the performance limits of the current approach.

5.3 Future research recommendations

We believe our implementation of the radial distribution function (RDF) has room for improvement. While it successfully identifies frequently occurring distances, it exhibits several shortcomings, most notably in scaling. Scaling the histograms would better represent the actual probabilities involved and enhance the visibility of the peaks in the plot. Additionally, periodic boundary conditions (PBC) were not accounted for, resulting in a plot with a limited domain that does not accurately reflect the modeled system. Another issue lies in the "histogram-over-time" approach. While conceptually appealing, it did not yield meaningful insights. A better alternative might have been to compute a RDF that is averaged over increasing time intervals, to illustrate the system's expected change in behavior. This would likely show the distribution converging into a smooth curve as equilibrium is reached, providing a clearer picture of the system's dynamics.

Another important aspect for future research is the accuracy of the physical model, particularly for larger molecular systems. While our implementation is generalized, it may not fully capture the behavior of molecules with long intramolecular distances. Our current approach includes three intramolecular forces, which effectively model molecules where the longest atomic separation is at most three bonds. However, for molecules with longer bond distances, further investigation is required to assess whether the Lennard-Jones potential remains sufficiently accurate. If the Lennard-Jones forces appear to model molecules with big intramolecular distances in an accurate manner, the current implementation should suffice.

The system's initialization step also has certain flaws. After creating the lattice configuration, we perform a brief simulation with bounded force magnitudes to separate the molecules and prevent numerical errors during integration. Although this approach works in practice, it is not physically accurate, introduces unnecessary complexity to the code structure, and fails to address the root issue. One potential solution is to adjust the grid placement algorithm to exclude atoms without Lennard-Jones (LJ) interactions. For example, in the case of a water molecule, rather than placing the molecule's center point on the lattice,

only the oxygen atom would be placed. Since the hydrogen atoms do not participate in LJ interactions, this adjustment would maximize distances between LJ-interacting pairs and (ideally) mitigate numerical instability during integration.

Finally, one last interesting idea is to remove redundancy from the model inputs, and only work with the ITP files as input data. Since these ITP files contain all information needed to construct a molecule (bond lengths and angles), it would be an interesting challenge to construct an initial configuration from only this information. While this approach would not impact the simulation results, it presents an engaging technical challenge for those interested.

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A Appendix

A.1 Symbols and units

This report uses a lot of symbols with non-standard units. Table A.1 shows an overview of all symbols with its unit.

Symbol	Unit	Meaning
H	-	Hydrogen atom
O	-	Oxygen atom
C	-	Carbon atom
$\mathbf{F}_i$	kg nm/ps^2	Force acting on atom i
m_i	kg	Mass of particle i
$\mathbf{q}_i$	nm	Position of particle i
$\dot{\mathbf{q}}_i = \mathbf{v}_i$	nm/ps	Velocity of particle i
$\ddot{\mathbf{q}}_i = \mathbf{a}_i$	nm/ps^2	Acceleration of particle i
V_{bond}	kJ/mol	Bond potential
V_{angle}	kJ/mol	Angular potential
V_{dihedral}	kJ/mol	Dihedral potential
V_{pp}	kJ/mol	Particle-particle potential energy
$\mathbf{r}_{ij}$	nm	Distance vector from atom i to atom j
r_{ij}	nm	Distance between atom i and atom j
r_0	nm	Most energetically favorite distance between two atoms
θ_{ijk}	rad	The angle between i, j and k
θ_0	rad	Most energetically favorite angle between three atoms
k_b	kJ/(mol nm^2)	Bond strength
k_θ	kJ/(mol rad^2)	Angle strength
U_{ij}^{LJ}	kJ/mol	Lennard Jones interaction potential
σ_i	nm	The distance at which V_{pp} is zero for particle i
ϵ_{ij}	kJ/mol	$V(2^{1/6}r) = -\epsilon$
θ_{ijkl}	rad	The angle between the plane through atoms i, j and k and the plane through atoms j, k and l

Table A.1: Meaning of all symbols with unit if applicable

A.2 Intramolecular forces

In this section, we look into the different forces that are affecting the acceleration of each atom within a molecule. There are three forces taken into account in this report:

- the bond force;
- the angular force;
- the dihedral force.

We look at the force between different molecules in the next section.

A.2.1 Bond force

The magnitude of the bond force $|\mathbf{F}_{\text{bond},ij}|$ on atom i caused by atom j, can be determined by calculating the derivative to r of the bond potential:

$$|\mathbf{F}_{\text{bond},ij}| = -\left.\frac{\partial V_{\text{bond}}(r)}{\partial r}\right|_{r=r_{ij}} = k_b(r_0 - r_{ij}) \tag{A.2.1}$$

The direction of $\mathbf{F}_{\text{bond},ij}$ is the same as $\mathbf{r}_{ij}$ if $r_{ij} > r_0$ and the opposite direction otherwise. The first case is illustrated in Figure 2.2. Note that k_b is still depending on which atoms i and j are interacting. We know from Newton's third law that:

$$\mathbf{F}_{\text{bond},ij} = -\mathbf{F}_{\text{bond},ji} \tag{A.2.2}$$

The total bond force $\mathbf{F}_{\text{bond},i}$ on atom i is now equal to:

$$\mathbf{F}_{\text{bond},i} = \sum_{\text{bonded } j} \mathbf{F}_{\text{bond},ij} = \sum_{\text{bonded } j} |\mathbf{F}_{\text{bond},ij}| \cdot \frac{-\mathbf{r}_{ij}}{|\mathbf{r}_{ij}|} = \sum_{\text{bonded } j} k_b(r_{ij} - r_0) \cdot \frac{\mathbf{r}_{ij}}{|\mathbf{r}_{ij}|} \tag{A.2.3}$$

A.2.2 Angular force

A.2.2.1 Angular force on i and k

We look at the angular force on i. By symmetry, the angular force on k can be found completely analogously.

The magnitude of the force $|\mathbf{F}_{\text{angle},i,ijk}|$ can be determined by calculating the derivative to r of the angle potential. For this we use the equation $r\partial r = \sin\partial\theta \approx \partial\theta$:

$$|\mathbf{F}_{\text{angle},i,ijk}| = -\left.\frac{\partial V_{\text{angle}}(\theta)}{\partial r}\right|_{\theta=\theta_{ijk}} = -\frac{1}{r}\left.\frac{\partial V_{\text{angle}}(\theta)}{\partial \theta}\right|_{\theta=\theta_{ijk}} = \frac{k_b}{|\mathbf{r}_{ij}|}(\theta_0 - \theta_{ijk}) \tag{A.2.4}$$

The direction of $\mathbf{F}_{\text{angle},i,ijk}$ is perpendicular to $\mathbf{r}_{ij}$ in the ijk-plane. If $\theta_{ijk} > \theta_0$, the direction is the same as in Figure 2.4 and the opposite direction otherwise. This direction can be found using a trick: We find $\hat{\mathbf{n}}_{ijk}$ which is the unit vector that is perpendicular to the ijk-plane, in the following way:

$$\hat{\mathbf{n}}_{ijk} = \frac{\mathbf{r}_{ji} \times \mathbf{r}_{jk}}{|\mathbf{r}_{ji} \times \mathbf{r}_{jk}|} \tag{A.2.5}$$

We can then find the direction of $\mathbf{F}_{\text{angle},i,ijk}$ with the distance vector from i to j to find the final form:

$$\mathbf{F}_{\text{angle},i,ijk} = |\mathbf{F}_{\text{angle},i,ijk}| \frac{\mathbf{r}_{ij} \times \hat{\mathbf{n}}_{ijk}}{|\mathbf{r}_{ij} \times \hat{\mathbf{n}}_{ijk}|} = \frac{k_b}{|\mathbf{r}_{ij}|}(\theta_0 - \theta_{ijk}) \frac{\mathbf{r}_{ij} \times \hat{\mathbf{n}}_{ijk}}{|\mathbf{r}_{ij} \times \hat{\mathbf{n}}_{ijk}|} \tag{A.2.6}$$

Note that $|\mathbf{F}_{\text{angle},i,ijk}|$ is not necessarily the same as $|\mathbf{F}_{\text{angle},k,ijk}|$ as the magnitude of the force not only depends on the angle, but also the distance from i to j or k to j respectively. Note that in this definition:

$$\mathbf{F}_{\text{angle},i,ijk} = \mathbf{F}_{\text{angle},i,kji} \tag{A.2.7}$$

A.2.2.2 Angular force on j

The sum of the forces should be zero, so we can find that $\mathbf{F}_{\text{angle},j,ijk}$ is determined as follows:

$$\mathbf{F}_{\text{angle},j,ijk} = -\left(\mathbf{F}_{\text{angle},i,ijk} + \mathbf{F}_{\text{angle},k,ijk}\right) \tag{A.2.8}$$

The total angle force $\mathbf{F}_{\text{angle},i}$ on atom i is now equal to:

$$\mathbf{F}_{\text{angle},i} = \sum_{\text{all triples } pqr \text{ containing } i} \mathbf{F}_{\text{angle},i,pqr} \tag{A.2.9}$$

where all triples can be found in Table B.3 for water and Table B.6 for ethanol.

A.2.3 Dihedral force

A.2.3.1 Dihedral force on i and l

We look at the dihedral force on i. By symmetry, the dihedral force on l can be found completely analogously.

The magnitude of the force $|\mathbf{F}_{\text{dihedral},i,ijkl}|$ can be determined by calculating the derivative to r of the angle potential. For this we use the equation $r\sin\theta_{ijk}\partial r = \sin\partial\psi \approx \partial\psi$:

$$\begin{aligned}|\mathbf{F}_{\text{dihedral},i,ijkl}| &= -\frac{\partial V_{\text{dihedral}}(\psi)}{\partial r} = -\frac{1}{r}\frac{\partial V_{\text{dihedral}}(\psi)}{\partial \psi} \\ &= \frac{1}{2|\mathbf{r}_{ij}|\sin\theta_{ijk}}\Big[C_1\sin\psi - 2C_2\sin 2\psi + 3C_3\sin 3\psi - 4C_4\sin 4\psi\Big] \end{aligned} \tag{A.2.10}$$

The direction of $\mathbf{F}_{\text{dihedral},i,ijkl}$ is perpendicular to the ijk-plane or parallel to $\hat{n}_{ijk}$ as defined in Equation A.2.5. We assume a positive force is in the direction as in Figure 2.5 and the opposite direction otherwise. We find the following expression for the force:

$$\begin{aligned}\mathbf{F}_{\text{dihedral},i,ijkl} &= |\mathbf{F}_{\text{dihedral},i,ijkl}|\,\hat{\mathbf{n}}_{ijk} \\ &= \frac{1}{2|\mathbf{r}_{ij}|\sin\theta_{ijk}}\Big[C_1\sin\psi - 2C_2\sin 2\psi + 3C_3\sin 3\psi - 4C_4\sin 4\psi\Big]\hat{\mathbf{n}}_{ijk} \end{aligned} \tag{A.2.11}$$

Note that in this definition:

$$\mathbf{F}_{\text{dihedral},i,ijkl} = \mathbf{F}_{\text{dihedral},i,lkji} \tag{A.2.12}$$

A.2.3.2 Dihedral force on j and k

The calculations of this paragraph are based on section 4.2 of a paper by Monasse and Boussinot [14]. We define the point o_{ijkl} to be the center between j and k. We can find it using $j + (\mathbf{r}_{jk})/2$. We know that the sum of both the forces and the torques have to be zero:

$$\sum_{\alpha\in\{i,j,k,l\}} \mathbf{F}_{\text{dihedral},\alpha,ijkl} = 0 \qquad \sum_{\alpha\in\{i,j,k,l\}} \mathbf{r}_{o\alpha}\times\mathbf{F}_{\text{dihedral},\alpha,ijkl} = 0 \tag{A.2.13}$$

We find the following expression with some calculations (details in Monasse and Boussinot [14]):

$$\begin{aligned}\mathbf{r}_{ok}\times\mathbf{F}_{\text{dihedral},k,ijkl} &= -\left(\mathbf{r}_{ok}\times\mathbf{F}_{\text{dihedral},l,ijkl} + \frac{1}{2}\mathbf{r}_{kl}\times\mathbf{F}_{\text{dihedral},l,ijkl} + \frac{1}{2}\mathbf{r}_{ji}\times\mathbf{F}_{\text{dihedral},i,ijkl}\right) \\ &:= \mathbf{t}_k \end{aligned} \tag{A.2.14}$$

Since we have infinite solutions, we can choose one:

$$\mathbf{F}_{\text{dihedral},k,ijkl} = \left(\frac{1}{|\mathbf{r}_{ol}|^2}\right)\mathbf{t}_k\times\mathbf{r}_{ok} \tag{A.2.15}$$

We can now find $\mathbf{F}_{\text{dihedral},j,ijkl}$ by using the fact that the forces have to add up to zero (Equation A.2.13):

$$\mathbf{F}_{\text{dihedral},j,ijkl} = -\mathbf{F}_{\text{dihedral},i,ijkl} - \mathbf{F}_{\text{dihedral},k,ijkl} - \mathbf{F}_{\text{dihedral},l,ijkl} \tag{A.2.16}$$

The total dihedral force on an atom i is now determined by:

$$\mathbf{F}_{\text{dihedral},i} = \sum_{\text{all quadruples } pqrs \text{ containing } i} \mathbf{F}_{\text{dihedral},i,pqrs} \tag{A.2.17}$$

where all quadruples can be found in Table B.7.

A.3 Intermolecular forces

The magnitude of the Lennard Jones force $|\mathbf{F}_{\mathrm{LJ},ij}|$ on atom i caused by atom j, can be determined by calculating the derivative to r of the Lennard Jones potential:

$$|\mathbf{F}_{\mathrm{LJ},ij}| = -\left.\frac{\partial U_{ij}^{\mathrm{LJ}}(r)}{\partial r}\right|_{r=r_{ij}} = \frac{24\epsilon_{ij}}{r_{ij}}\left[2\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] \tag{A.3.1}$$

The direction of $\mathbf{F}_{\mathrm{LJ},ij}$ is the same as in Figure 2.6 if the force is positive and the opposite direction otherwise. Therefore we find that:

$$\mathbf{F}_{\mathrm{LJ},ij} = |\mathbf{F}_{\mathrm{LJ},ij}| \cdot \frac{\mathbf{r}_{ij}}{|\mathbf{r}_{ij}|} = \frac{24\epsilon_{ij}}{r_{ij}}\left[2\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] \cdot \frac{\mathbf{r}_{ij}}{|\mathbf{r}_{ij}|} \tag{A.3.2}$$

We know from Newton's third law that:

$$\mathbf{F}_{\mathrm{LJ},ij} = -\mathbf{F}_{\mathrm{LJ},ji} \tag{A.3.3}$$

The total Lennard Jones force $\mathbf{F}_{\mathrm{LJ},i}$ on atom i is now equal to:

$$\mathbf{F}_{\mathrm{LJ},i} = \sum_{\text{non-bonded } j} \mathbf{F}_{\mathrm{LJ},ij} = \sum_{\text{non-bonded } j} \frac{24\epsilon_{ij}}{r_{ij}}\left[2\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] \cdot \frac{\mathbf{r}_{ij}}{|\mathbf{r}_{ij}|} \tag{A.3.4}$$

B Parameters

B.1 Water

The bonds and angles listed below use the indices in Figure 2.1a. The parameters below were taken from the TIP3P water model (taken from [15], but also found in [9]).

Species	Mass (amu)	σ (nm)	ϵ (kJ/mol)
H	1.0080	0	0
O	15.9994	0.315061	0.66386

Table B.1: H_2O masses and Lennard Jones parameters.

ij	R_0 (nm)	k_b (kJ mol^{-1} nm^{-2})
01,02	0.09572	502416

Table B.2: H_2O bonds.

ijk	θ_0 (°)	k_θ (kJ mol^{-1} rad^{-2})
102	104.52	628.02

Table B.3: H_2O angles.

B.2 Ethanol

The bonds and angles listed below use the indices in Figure 2.1b. The parameters below were taken from the OPLS/AA force-field (taken from [15], but also found in [10]).

Species	Mass (amu)	σ (nm)	ϵ (kJ/mol)
H (bonded to **C**)	1.0080	0.25	0.125520
H (bonded to **O**)	1.0080	0	0
C	12.0110	0.35	0.276144
O	15.9994	0.312	0.711280

Table B.4: Ethanol masses and Lennard Jones parameters.

ij	R_0 (nm)	k_b (kJ mol^{-1} nm^{-2})
04	0.1529	224262.4
20,30,10,54,64	0.1090	284512.0
47	0.1410	267776.0
87	0.0945	462750.0

Table B.5: Ethanol bonds.

ijk	θ_0 (°)	k_θ (kJ mol^{-1} rad^{-2})
104,204,304	108.5	292.880
302,301,201,546	107.8	276.144
046,045	110.7	313.800
047	109.5	414.400
478	108.5	460.240
547,647	109.5	292.880

Table B.6: Ethanol angles.

$ijkl$	C_1 (kJ mol^{-1})	C_2 (kJ mol^{-1})	C_3 (kJ mol^{-1})	C_4 (kJ mol^{-1})
1045,2045,3045,1046,2046,3046	0.62760	1.88280	0.00000	-3.91622
1047,2047,3047	0.97905	2.93716	0.00000	-3.91622
0478	-0.44310	3.83255	0.72801	-4.11705
5478,6478	0.94140	2.82420	0.00000	-3.76560

Table B.7: Ethanol dihedrals.

C Code

C.1 Integrators

C.1.1 Euler

The following code is explained in 3.2.1.1.

```
euler = {
    'positions': lambda x_im1, x_im2, v_im1, a_im1, a_im2, dt:
                    x_im1 + dt * v_im1 + (dt**2) / 2 * a_im1,
    'velocities': lambda x_i, x_im1, x_im2, v_im1, a_i, a_im1, a_im2, dt:
                    v_im1 + dt * a_im1
}
```

C.1.2 Verlet

The following code is explained in 3.2.1.2.

```
verlet = {
    'positions': lambda x_im1, x_im2, v_im1, a_im1, a_im2, dt:
                    2 * x_im1 - x_im2 + ((dt**2) * a_im1),
    'velocities': lambda x_i, x_im1, x_im2, v_i, a_i, a_im1, a_im2, dt:
                    1 / (2 * dt) * (x_i - x_im2)
}
```

C.1.3 Velocity Verlet

The following code is explained in 3.2.1.3.

```
velo_verlet = {
    'positions': lambda x_im1, x_im2, v_im1, a_im1, a_im2, dt:
                    x_im1 + dt * v_im1 + ((dt**2) / 2 * a_im1),
    'velocities': lambda x_i, x_im1, x_im2, v_i, a_i, a_im1, a_im2, dt:
                    v_im1 + dt / 2 * (a_im1 + a_i)
}
```

C.1.4 Beeman

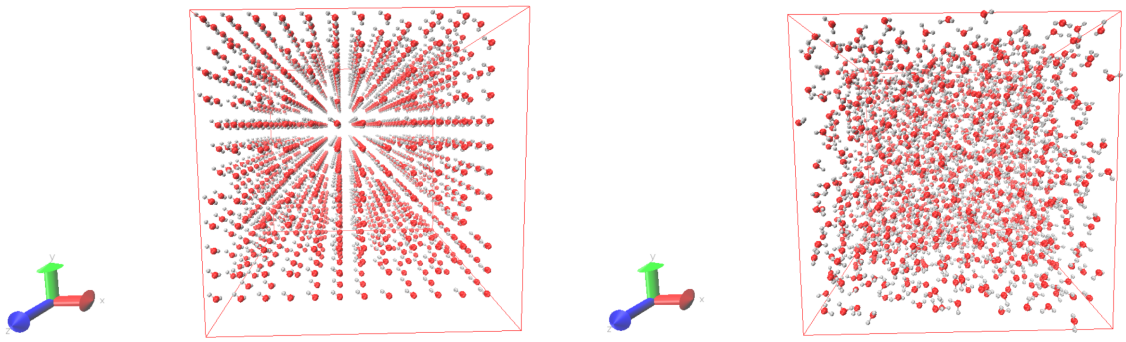
The following code is explained in 3.2.1.4.

```
beeman = {
    'positions': lambda x_im1, x_im2, v_im1, a_im1, a_im2, dt:
                    x_im1 + dt * v_im1 + (4*a_im1-a_im2)*(1/6)*dt**2,
    'velocities': lambda x_i, x_im1, x_im2, v_i, a_i, a_im1, a_im2, dt:
                    v_im1 + (2*a_i+5*a_im1-a_im2)*dt/6
}
```

D Results

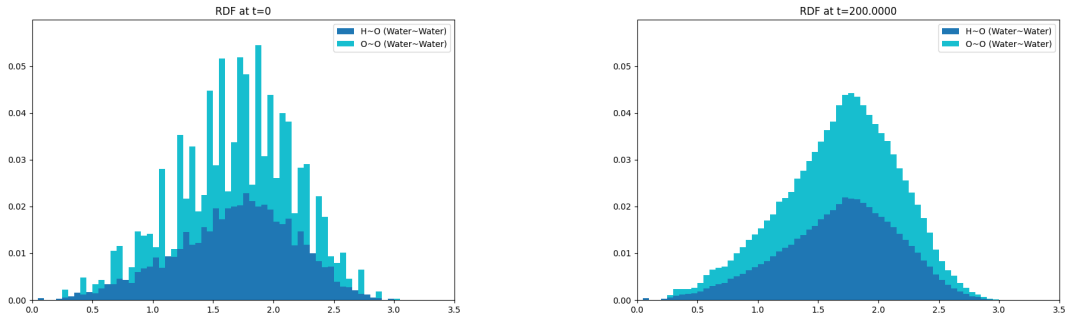
D.1 Pure water system

For this system, we let `weights=[1,0,0,0]`. Running the simulation provides the following figures:



(a) Initial configuration (b) Later configuration

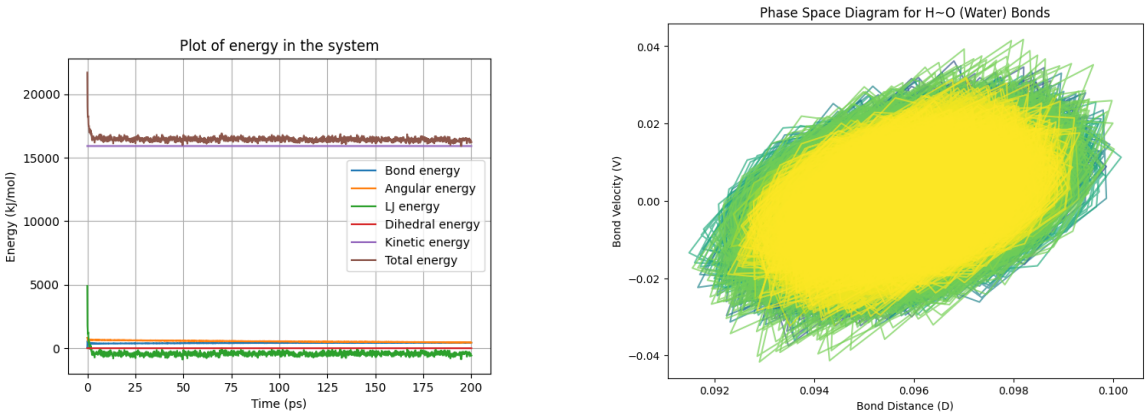
Figure D.1: Two frames of the simulation, visualized with VMD



(a) at $t = 0$ ps

(b) at $t = 200$ ps

Figure D.2: Radial distribution of the **H**-**O** and **O**-**O** bonds. x-axis is in nm.



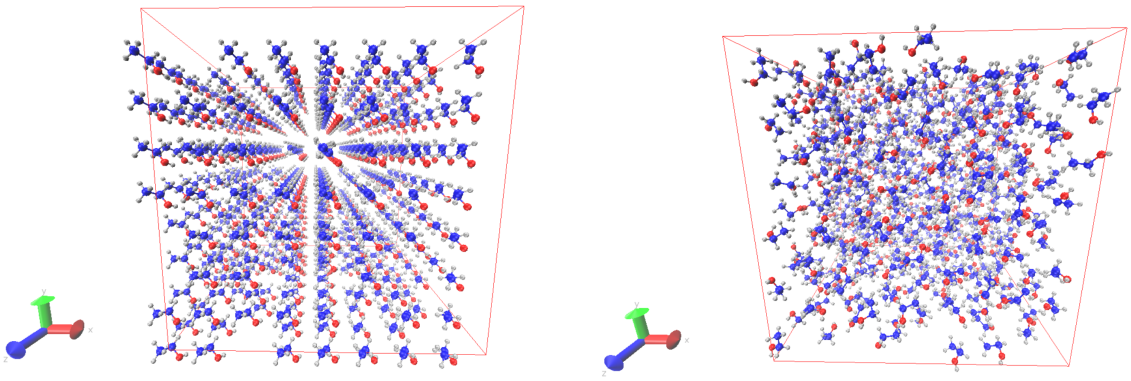
(a) Energy of the system over time

(b) Phase space diagrom for **H**-**O** bonds

Figure D.3: Two frames of the simulation, visualized with VMD

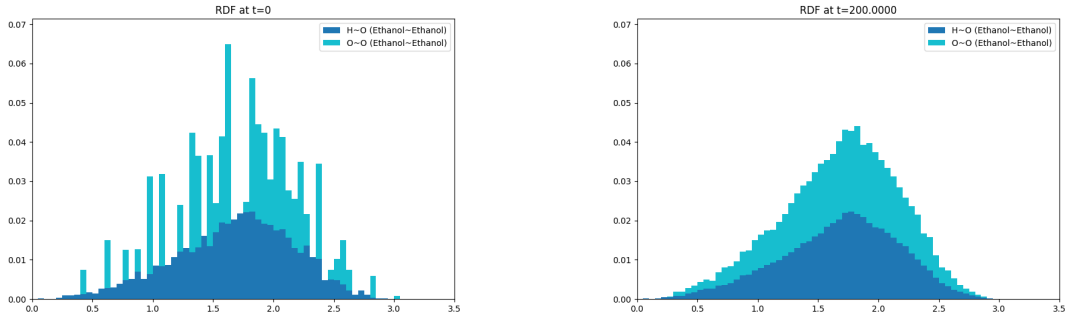
D.2 Pure ethanol system

For this system, we let `weights=[0,1,0,0]`. Running the simulation provides the following figures:



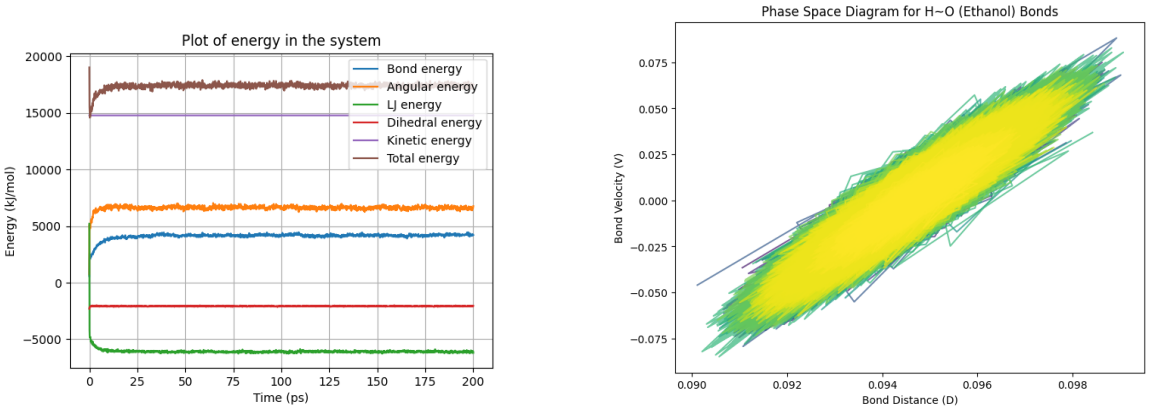
(a) Initial configuration (b) Later configuration

Figure D.4: Two frames of the simulation, visualized with VMD



(a) at $t = 0$ ps (b) at $t = 200$ ps

Figure D.5: Radial distribution of the **H**-**O** and **O**-**O** bonds. x-axis is in nm.



(a) Energy of the system over time (b) Phase space diagrom for **H**-**O** bonds

Figure D.6: Two frames of the simulation, visualized with VMD